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SK 360°: A CENTRALIZED DIGITAL PLATFORM FOR YOUTH GOVERNANCE IN LIPA CITY

A Capstone Project

Submitted to the Faculty of

NU Lipa - School of Architecture Computing and Engineering

In Partial Fulfillment of the Requirements for the Degree

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CHAPTER 1
INTRODUCTION

In the era of rapid digital transformation in the Philippines, the adoption of modern information and communication technologies (ICT) has become essential in improving transparency, coordination, and efficiency in local governance. National initiatives led by the Department of Information and Communications Technology (DICT), such as the eGovPH Super App and integrated digital governance frameworks, highlight the growing shift toward centralized and technology-driven public service systems. Within this context, local governance institutions are encouraged to adopt digital platforms that support organized documentation, systematic reporting, and effective communication. Aligned with this direction, digital systems that integrate real-time communication, role-based access control, automated reporting, centralized document management, and basic data analytics play a crucial role in promoting efficient and transparent governance, particularly in youth-led institutions.

The Sangguniang Kabataan (SK) serves as the primary youth governance body in the Philippines, mandated to lead youth participation, strengthen civic engagement, and promote accountability in local communities as stated in the SK Reform Act of 2015 [1]. Youth governance plays an important role in democratic development because institutions like the SK help build young leaders, promote public participation, and support long-term community involvement. National agencies such as the Department of the Interior and Local Government (DILG) emphasize that effective SK operations rely on organized documentation, coordinated reporting structures, and consistent monitoring of youth programs across barangays [2]. With this, having clear, transparent, and systematic processes becomes essential to good governance.

Within cities and municipalities, the SK Federation acts as the central coordinating body that connects all barangay SK councils. It is responsible for consolidating accomplishment reports, monitoring youth activities, issuing announcements, and guiding SK leaders in the implementation of community-based programs. However, many of these workflows are still done manually, relying on paper-based documentation and communication through informal channels such as group chats [3], [4]. These fragmented practices cause late submissions, inconsistent file formats, data loss, and difficulty maintaining organized and

auditable records, especially when compiling monthly, quarterly, and annual accomplishment reports that require uniformity and proper tracking.

These challenges are more evident in Lipa City, which has seventy-two barangays with different administrative capacities and digital practices. The Federation often experiences delays, incomplete submissions, inconsistent report formats, and difficulty consolidating documents across barangays. Per existing policy, financial validation remains the sole responsibility of the barangay SK Chairman prior to submission; the Federation and the proposed system perform no auditing function. This means the Federation's role is limited to receiving, recording, consolidating, and archiving reports without performing financial verification to follow existing administrative procedures. Leadership transitions every three years also create challenges because newly elected SK officials must gain access to previous reports and records. Without a centralized repository, documents become scattered, lost, or inaccessible, making it harder for new councils to continue existing programs.

Studies on digital governance show that local government units can improve efficiency, coordination, and transparency by adopting ICT tools that support existing workflows [5], [6]. While some LGUs already use digital platforms, most are not designed specifically for SK governance and often lack youth-focused features such as standardized accomplishment report formats, organized leadership profiles, centralized communication, and support for leadership transitions [7], [8]. This reveals a technological gap in digital tools made specifically for youth-governance processes in the Philippines.

To address these challenges, this study proposes SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City. The system integrates essential SK functions, including accomplishment report submission, structured monthly, quarterly, and annual report compilation, budget document uploads, leadership profile management, announcements, meeting schedules, and an event calendar accessible to all SK accounts. Role-based access is implemented for the SK Federation President/Admin, SK Chairman, and SK Secretary to ensure secure and appropriate system use. Communication is enhanced through real-time chat and video call/video chat features, while a gamified ranking system encourages participation based on timely submissions, engagement, and program completion. The platform also includes basic data analytics to provide insights on submission trends, program effectiveness, and youth engagement. While all kabataan in

Lipa City may access public features such as announcements, calendars, and leadership profiles, advanced functions like report submission and document uploads are restricted to authorized SK officials. The system further supports leadership transitions by preserving historical records and leadership information, promoting organized, transparent, and efficient youth governance aligned with national guidelines and the operational needs of the SK Federation of Lipa City.

Overall, this study contributes to the modernization of youth governance through a digital system built specifically for SK workflows. While other LGU platforms focus on general administrative needs, this system is designed for youth-governance requirements such as report standardization, inter-barangay coordination, communication, and leadership transition support.

1.1 PROJECT CONTEXT

The SK Federation of Lipa City oversees the activities, reports, and coordination of the seventy-two barangay SK councils in the city, as mandated by the SK Reform Act of 2015 [1]. Despite this important responsibility, the current workflow remains largely manual and fragmented. Barangay SK councils submit physical documents, use inconsistent digital formats, and rely on group chats for communication, which often result in delayed submissions, incomplete records, duplicated files, and difficulties in organizing citywide documentation [3], [4]. Variations in the digital capabilities of barangays further make coordination more challenging for the Federation.

The proposed system aims to centralize essential SK functions, including accomplishment report submission and compilation, budget document uploads, announcement posting, meeting scheduling, leadership profile management, real-time messaging, video call for virtual meetings, a gamified ranking system to encourage timely and quality submissions, and data analytics for tracking performance metrics. A role-based access structure is implemented for the SK Federation President/Admin, SK Chairman, and SK Secretary to ensure organized and secure system usage. The platform also includes an event calendar accessible to all SK accounts to improve scheduling and coordination. To promote broader youth involvement, Kabataan in Lipa City are provided limited access to public information such as announcements, calendars, and rankings, while sensitive system functions remain restricted to authorized SK officials.

The system follows existing administrative procedures, wherein the financial validation of budget reports is handled solely by the barangay SK Chairman prior to submission. The platform does not perform budget verification but serves as a centralized repository for uploading, storing, and archiving documents, supported by analytics dashboards for non-financial insights. It also preserves historical records to support leadership transitions, ensuring that newly elected SK officials can access previous submissions and documentation. This addresses common concerns raised by the Commission on Audit (COA) and the Department of the Interior and Local Government (DILG) regarding record loss and weak documentation during leadership turnover.

This project supports Sustainable Development Goal (SDG) 16, which promotes transparent and accountable institutions, and SDG 17, which emphasizes strengthened coordination and partnerships [9], [10]. By offering a digital platform designed specifically for SK operations, the study addresses a significant gap in existing youth governance systems. Unlike general LGU platforms, the proposed system focuses on SK-specific reporting and coordination needs, making it valuable for improving documentation, communication, and continuity within the SK Federation of Lipa City.

1.2 OBJECTIVES OF THE STUDY

General Objective

The general objective of this study is to design and develop SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City that supports accomplishment reporting, budget documentation, communication, gamified engagement, data analytics, and secure access for the SK Federation and barangay SK councils in Lipa City, while allowing limited youth access to public information, and to evaluate the system based on functionality, usability, reliability, performance efficiency, and security.

Specific Objectives

Specifically, this study aims to:

1. To design and develop a system that enables SK officials to submit reports, upload documents, and access system features efficiently, while allowing Kabataan limited access to public content.

2. To develop a structured reporting module for automated monthly, quarterly, and annual accomplishment reports in PDF format.

3. To implement secure role-based access control for the SK Federation President/ Admin, SK Chairman, and SK Secretary.

4. To establish centralized announcements, an integrated event calendar, real-time chat, video call, gamified rankings, and basic data analytics to support coordination and engagement.

5. To evaluate the developed system based on the ISO/IEC 25010 software quality attributes functionality, usability, reliability, performance efficiency, and security.

1.3 SCOPE AND DELIMITATION

Scope of the Project

This study focuses on the design and development of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City, which supports the actual operational processes of the Sangguniang Kabataan. The system covers the digital submission of barangay SK accomplishment reports, wherein SK Secretaries upload reports that are reviewed and consolidated by the SK Federation for monthly, quarterly, and annual documentation. Budget-related documents are uploaded by authorized barangay SK officials for record-keeping, while financial validation remains the responsibility of the SK Chairman at the barangay level. The platform also supports common SK activities such as posting announcements, scheduling meetings, coordinating events through an integrated calendar, and communicating through real-time chat and video call/video chat features. Leadership profile management and document archiving are included to preserve records and ensure continuity during leadership transitions. Role-based access is implemented for the SK Federation President/Admin, SK Chairman, and SK Secretary, while Kabataan are provided limited access to public information such as announcements, calendars, and rankings. The system is developed exclusively for the SK Federation of Lipa City and is limited to the defined features within the academic research timeframe.

The platform emphasizes federation-level consolidation to enable efficient oversight and data-driven decision-making across all barangays in Lipa City. By incorporating gamified

rankings and mobile accessibility, it aims to boost youth motivation and participation in SK activities. Overall, SK 360° serves as a practical tool to strengthen transparency, coordination, and institutional memory in local youth governance.

Delimitation of the Project

This study is limited to the development of a web and mobile application designed exclusively for the SK Federation of Lipa City. While the system includes core modules such as accomplishment report submission, automated monthly, quarterly, and annual report compilation, budget documentation uploads, announcements, meeting schedules, leadership profile management, an integrated event calendar, and real-time communication features, it does not cover advanced functionalities such as automated budget auditing, comprehensive performance analytics, project evaluation tools, or integration with external LGU and COA databases. Financial verification of budget reports is excluded from the system, as this responsibility remains with the barangay SK Chairman prior to submission.

The study also excludes system expansion for other municipalities or LGU departments, inter-city or nationwide SK coordination, online payment gateways, SMS broadcasting systems, advanced analytics beyond basic insights, offline system operation, automated election modules, and extensive customization options. The development is limited to registered SK officials and authorized users within Lipa City and is conducted within the academic research timeframe. The system functions solely as an internal digital platform for coordination, documentation, and record-keeping and does not replace the mandatory hard-copy submissions required by the DILG or COA, ensuring continued compliance with existing regulatory requirements.

These delimitations keep the project focused, feasible, and aligned with academic constraints while allowing for targeted validation in Lipa City's context. Future expansions could incorporate excluded features based on user feedback and operational success.

CHAPTER 2
REVIEW OF RELATED LITERATURE, STUDIES AND SYSTEMS

2.1 Related Literatures and Studies

This section presents a thematic discussion of existing foreign and local literatures and studies related to youth governance, Sangguniang Kabataan (SK) operations, the use of digital systems in local governance, and the application of engagement technologies such as gamification and real-time communication tools. The review highlights recurring issues, theoretical perspectives, and technological approaches that are relevant to the proposed project.

Youth Governance and the Sangguniang Kabataan

Republic Act No. 10742, also known as the Sangguniang Kabataan Reform Act of 2015 [1], was enacted to strengthen youth participation, transparency, and accountability in local governance. While the law granted fiscal autonomy and mandated leadership training for SK officials, several authors emphasize that operational challenges remain, particularly in documentation, coordination, and reporting processes. Studies indicate that the effectiveness of youth councils is strongly influenced by administrative capacity, leadership continuity, and access to organized digital systems that support governance functions beyond traditional manual practices. These findings suggest that legal reforms alone are insufficient without parallel technological support mechanisms.

Challenges in SK Operations and Documentations

Local studies reveal that many SK councils still rely on manual and semi-digital methods such as paper-based records and communication through social media group chats. Rosales [3] found that inconsistent documentation practices and delayed submission of accomplishment reports negatively affect program planning, monitoring, and evaluation. Similarly, Padillo and Vallespin [4] reported that fragmented communication and the absence of standardized reporting tools contribute to low perceptions of SK effectiveness and reduced youth participation. These findings suggest that procedural inefficiencies weaken the intended outcomes of the SK Reform Act.

Further, Flores et al. [11] emphasized that leadership transitions every three years often result in the loss of institutional memory due to poor record-keeping and the absence of centralized repositories. Damit [8] also highlighted that despite expanded mandates, SK councils remain constrained by limited organizational capacity and weak monitoring mechanisms. More recent assessments, such as multi-sectoral evaluations in Taguig City [14] and qualitative studies on advocacy organization in Pagadian City (Lastimado et al. [13]), continue to highlight persistent concerns over accountability, logistical barriers, resource constraints, and coordination issues in SK operations, reinforcing the need for structured digital support in documentation, consolidation, and inter-barangay collaboration. These studies collectively underscore the need for digital support systems that can institutionalize processes, preserve historical records, and ensure continuity across leadership terms.

Digital Governance and ICT in Local Government

Foreign and local literature on e-governance supports the adoption of information and communication technologies to improve transparency, efficiency, and coordination in local governments. Siar [6] observed that early LGU websites in the Philippines provided limited interactivity and failed to support operational workflows, focusing mainly on information dissemination. More recent frameworks from the United Nations Development Programme [5] emphasize that integrated digital platforms can enhance coordination, crisis response, and sustainable development when aligned with existing governance structures. Systems like Kabataan-Konek [15], a barangay-level web-based youth information management tool with data analytics, demonstrate progress in youth profiling and basic monitoring but remain limited to single-barangay scopes without federation-level integration or advanced engagement features. This limitation highlights the gap between isolated digital initiatives and scalable, city-level youth governance platforms.

Post-2020 initiatives underscore the potential of digital platforms for youth empowerment in the Philippines. For example, the Skill Our Future platform, launched by UNDP in partnership with Microsoft and others [18], offers free online courses on digital literacy and employability skills to bridge the digital divide among underserved youth. While not governance-focused, it complements SK tools by building digital readiness for civic participation, aligning with the need for accessible platforms in youth-led institutions like the proposed SK 360°

Gamification in Digital Governance and Youth Engagement

Recent foreign and local studies highlight the role of gamification in increasing user motivation, participation, and task compliance in digital platforms. Gamification refers to the use of game-like elements such as points, rankings, leaderboards, and performance indicators in non-game environments. Research shows that these elements are particularly effective in youth-oriented systems, as they encourage healthy competition, accountability, and sustained engagement [12]. In governance-related applications, gamification has been found to improve timely submission of reports and active participation without relying on monetary incentives, with systematic reviews indicating positive outcomes in motivation, civic learning, and enjoyment in e-participation platforms. In the context of SK operations, gamified ranking systems can motivate barangay SK councils to submit reports on time, actively participate in programs, and maintain consistent engagement, aligning with national youth survey findings on the need for motivational tools to boost sustained involvement [7].

Real-Time Communication and Video Conferencing in Governance Systems

The use of real-time communication tools such as instant messaging and video conferencing has become an essential component of modern digital governance systems. Studies indicate that these technologies improve coordination, reduce delays in decision-making, and minimize the need for frequent physical meetings. Video calls enable remote collaboration, especially for organizations managing multiple locations or jurisdictions. Integrated tools like chat and video conferencing are increasingly recommended for distributed governance bodies to overcome logistical barriers, as seen in post-pandemic adaptations by Philippine local government units (LGUs), where platforms facilitated continued public service delivery, virtual consultations, and crisis communication amid movement restrictions. These technologies are particularly relevant for rural and geographically dispersed barangays, where transportation costs and connectivity challenges may hinder frequent in-person coordination. For youth governance bodies like the SK Federation, real-time chat and video call features support efficient coordination among barangays, facilitate virtual meetings, and ensure continuous communication despite time and logistical constraints.

Youth-focused studies further reinforce this need. The YouthLed Philippines and Social Weather Stations [7] national survey reported low trust and engagement among Filipino youth and recommended the use of digital platforms to increase accessibility, participation, and transparency in youth institutions. Collectively, these literatures and studies establish that manual SK workflows, weak digital infrastructure, and lack of engagement mechanisms hinder governance performance and youth involvement.

Data Privacy Considerations in Youth Governance Platforms

A critical consideration in developing digital platforms for youth governance is compliance with data privacy regulations, particularly in handling personal information of young users and officials, as in the Philippines, Republic Act No. 10173 (Data Privacy Act of 2012) [16] mandates safeguards for personal data processed by government entities, including LGUs, requiring transparency, legitimate purpose, proportionality, and rights such as access, correction, and erasure for data subjects, while imposing obligations on personal information controllers (e.g., government agencies or platform operators) to implement security measures, notify breaches, and ensure lawful processing, with penalties for unauthorized access or disclosure. For SK-focused platforms like the proposed SK 360° in Lipa City, implementing encryption, role-based access, and consent mechanisms is essential to protect sensitive data (e.g., leadership profiles, reports, and chat logs) while aligning with national standards and promoting secure, trustworthy youth engagement, which is particularly vital in a federated system involving multiple barangays, where real-time features like chat and video calls could involve sharing personal details.

In summary, both foreign and local sources consistently point to documentation inefficiencies, coordination gaps, leadership transition issues, and limited engagement in SK operations, while also emphasizing the potential of ICT solutions, gamification, and real-time communication tools to address these challenges. These insights directly inform the design, features, and long-term vision of the proposed system, ensuring that it is responsive, inclusive, and adaptable to evolving governance needs. The proposed SK 360° platform addresses these gaps by integrating automated reporting, federation-level consolidation, gamified engagement, real-time communication, and historical preservation into a privacy-compliant ecosystem tailored to Lipa City. This holistic approach overcomes fragmentation in the literature, positioning SK 360° as a scalable model to enhance youth participation, accountability, and sustainable continuity.

2.2 RELATED SYSTEMS

Feature(s)	SK Web Portal	SKIES	Kabataan - Konek	E-Task	SK 360°
Country	**Philippines**	**Philippines**	**Philippines**	**Philippines**	**Philippines**
Application Type	**Web**	**Web**	**Web**	**Web**	**Mobile and Web**
Accomplishment Report Automation	No	Partial	No	Partial	Yes
Budget Document Uploads	No	No	No	Yes	Yes
Role-Based Access Control	Limited	Yes	Limited	Yes	Yes
Announcements & Calendar	Yes	Limited	No	Yes	Yes
Real-Time Chat / Video Call	No	No	No	No	Yes
Gamified Ranking System	No	No	No	No	Yes
Data Analytics	No	Yes	Yes	Limited	Yes
Leadership History & Records	No	Partial	No	No	Yes
Federation-Level Consolidation	No	Partial	No	No	Yes

Table 1 Gap Analysis Table of Related Systems and Applications

Several existing systems and applications have been developed to support youth governance, SK operations, and barangay management. However, these systems vary in scope, functionality, and level of integration.

The SK Web Portal developed by the National Youth Commission serves as an official platform for SK registration, access to guidelines, and dissemination of information. While it promotes standardization, it lacks advanced features such as automated accomplishment report compilation, real-time communication, and analytics.

The SK Information and Evaluation System (SKIES) focus on performance monitoring and evaluation of SK programs. It provides data-driven insights but does not support comprehensive document management, federation-level consolidation, or engagement features such as gamification.

Kabataan-Konek is a web-based youth information management system designed for barangay-level youth profiling and analytics. Although useful for demographic analysis, it is limited to single-barangay implementation and does not address inter-barangay coordination or reporting workflows.

The e-TASK (Enhancing Transparency and Accountability of the Sangguniang Kabataan) system manages calendars, financial records, and project documentation. While it

addresses transparency concerns, it does not integrate real-time communication, automated multi-period reporting, or historical data preservation for leadership transitions.

Lastly, general Barangay Management Systems commonly developed in capstone projects focus on resident records, certificate issuance, and administrative services. These systems are not tailored to SK-specific needs and lack youth governance–oriented modules.

The proposed SK 360° is a centralized digital platform developed specifically for youth governance in Lipa City, integrating automated accomplishment report compilation, real-time chat and video calls, gamified ranking systems, comprehensive data analytics, budget document uploads, leadership history preservation, and federation-level data consolidation. Accessible via both mobile and web applications with robust role-based access control, announcements, calendars, and privacy safeguards aligned with RA 10173, it addresses the fragmentation of existing tools by providing a unified ecosystem that enhances coordination, engagement, transparency, accountability, and seamless institutional continuity across barangay, city, and federation levels.

The comparison shows that while existing systems address specific aspects of SK operations, none provide a fully integrated platform that combines reporting automation, federation-level consolidation, engagement mechanisms, and continuity support. Collectively, these tools improve isolated functions such as registration, evaluation, profiling, transparency, or basic administration but fall short in creating a unified ecosystem. This fragmentation perpetuates inefficiencies in data sharing, decision-making, and institutional memory across barangays and cities.

Internationally, initiatives like the EU's DEEP Linking Youth project [17] showcase e-participation platforms using AI dashboards and social listening to engage youth in policymaking. Though not replicable in the Philippine SK context due to scale differences, it underscores the value of inclusive digital tools, which aligns with SK 360°'s focus on federation-level consolidation and real-time youth engagement insights

2.3 Synthesis

The reviewed literatures, studies, and systems collectively demonstrate that despite policy reforms, SK operations continue to face persistent challenges in documentation, coordination, reporting efficiency, and leadership transition. Manual processes and

fragmented digital tools contribute to delays, data loss, and reduced institutional continuity. At the same time, studies on e-governance and youth participation consistently emphasize the role of ICT in improving transparency, accountability, and engagement. These challenges are compounded by the lack of scalable tools that preserve historical data and facilitate smooth handovers between SK batches. Addressing them requires platforms that prioritize both operational efficiency and long-term sustainability.

Existing systems related to SK and barangay governance address isolated needs such as profiling, evaluation, or transparency, but they lack comprehensive, SK-specific integration at the federation level. None fully support automated multi-period reporting, real-time communication, gamified engagement, and historical record preservation within a single platform. This gap results in duplicated efforts, inconsistent data quality, and missed opportunities for data-driven youth program improvements. A centralized solution could bridge these fragmented systems while aligning with national SK mandates.

In contrast, emerging Philippine tools like Skill Our Future [18] show how digital skills platforms can boost youth motivation and participation, while international examples such as the EU's DEEP project [17] illustrate scalable e-participation mechanisms using AI and social listening. However, these initiatives often overlook localized governance workflows and data privacy imperatives under RA 10173 [16]. Skill Our Future, developed in partnership with UNDP, provides free online courses to build employability and leadership skills among underserved youth, but its general focus on skills development does not directly address the administrative and federated coordination challenges specific to SK operations. Similarly, the DEEP project emphasizes inclusive tools for capturing underrepresented youth voices in policymaking, yet its broader scope limits direct applicability to Philippine SK contexts.

Based on this synthesis, the proposed project, SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City, is justified as a necessary and original solution that addresses documented gaps in youth governance workflows. By integrating reporting, communication, analytics, engagement, and continuity features aligned with SK mandates, while incorporating privacy safeguards, the study offers a localized yet scalable digital platform. Tailored to Lipa City's context, SK 360° enables federation-wide data consolidation, ensures compliance with data privacy laws, and supports more efficient, inclusive, and sustainable youth governance practices.

CHAPTER 3
TECHNICAL BACKGROUND

This chapter presents the detailed technologies to be used in the development of the proposed system. Also, a thorough discussion on the technical description of the proposed system and its components are described.

3.1 DETAILS OF THE TECHNOLOGY TO BE USED

This section discusses the technologies and resources essential to the development of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City. It covers the software, hardware, network, and peopleware components that support the system's functionality, security, and scalability. The selected technologies are compatible with web and mobile platforms and are designed to support SK operations such as report submission, document archiving, communication, scheduling, and basic data analytics. Development tools, frameworks, and database systems were utilized to ensure secure data handling, efficient system performance, and real-time communication. The involvement of system analysts, developers, and designers ensures that the system is properly designed, developed, and maintained to meet the technical and operational needs of youth governance.

Software/Hardware Requirements

Table 2 Software Requirements for Systems Development

Software Classification	Software Name
Integrated Development Environment (IDE)	Visual Studio Code – used as the primary code editor for developing the web and mobile application due to its flexibility, extension support, and compatibility with multiple programming languages.
Programming Languages and Frameworks	PHP 8.2 (server-side scripting), JavaScript ES6 (client-side scripting), HTML5 / CSS3 (web structure and styling), React.js 18.3.1 (web front-end framework), Laravel 10.x (web application framework), Flutter 3.24 (mobile application framework)
Database	MySQL 8.0 – used to store user accounts, barangay profiles, accomplishment reports, budget

	documents, announcements, chat records, rankings, and analytics data.
Third-Party Libraries	Firebase Cloud Messaging (2025) (for real-time notifications and announcements), Dyte.io API (2025) (for real-time chat, video conferencing, and virtual meetings), Bootstrap 5.3 (for responsive web interface design), Chart.js 4.x (for data analytics and graphical reporting dashboards), PHPMailer / Mailgun API (2025) (for system notifications and email alerts)
Others	XAMPP 8.2.4 (local development server for backend and database testing), Hostinger (2025) (for live deployment and web hosting), GitHub (2025) (for version control and collaborative code management)

Table 2 lists the software requirements for developing mobile and web applications

Visual Studio Code (Version 1.94, 2025)

Visual Studio Code is an Integrated Development Environment (IDE) used for writing, editing, and managing source code for web and mobile applications. It was used in the development of SK 360° because it supports all the programming languages required by the system, including PHP, JavaScript, HTML, CSS, and Dart for Flutter. Its debugging tools, extension support, and built-in version control help the development team efficiently build, test, and maintain both the web and mobile components of SK 360°, ensuring consistency and faster development throughout the project lifecycle.

PHP (Version 8.2)

PHP functions as the main server-side scripting language of the system. It handles backend operations such as user authentication, role-based access control, report processing, document management, and database transactions. PHP was used in SK 360° because it provides reliable server-side processing and secure data handling, which are essential for managing official SK records and ensuring proper system functionality.

JavaScript (Version ES6, 2025)

JavaScript ES6 is used to implement dynamic and interactive features within the web application. It enhances user experience by enabling real-time updates, form validation, asynchronous data handling, and interactive dashboards. JavaScript helps SK 360°

deliver smoother user interactions and faster responses, particularly during report submission, data viewing, and dashboard navigation.

HTML5 / CSS3 (Version 2025)

HTML5 and CSS3 are used to structure and design the web interface of the system. HTML5 defines the layout and content of the application pages, while CSS3 provides styling, responsiveness, and visual consistency. These technologies were used in SK 360° to ensure the system is accessible, visually organized, and compatible across various devices and web browsers.

React.js (Version 18.3.1, 2025)

React.js is utilized as the web front-end framework for building responsive and reusable user interface components. It supports efficient rendering of dashboards, report views, and analytics pages, providing smooth navigation and improved user interaction for SK officials and administrators. React.js was used in SK 360° to handle frequently updated data efficiently and enhance overall system performance.

Laravel (Version 10.x, 2025)

Laravel serves as the primary web application framework for backend development. It simplifies routing, authentication, database management, and API integration through its Model-View-Controller (MVC) architecture. Laravel was used in SK 360° to provide a structured, secure, and maintainable backend system suitable for handling governance-related processes and sensitive information.

Flutter (Version 3.24, 2025)

Flutter is used for developing the mobile version of the SK 360° system. It supports cross-platform deployment for Android and iOS, ensuring consistent performance and design. Flutter helps SK 360° extend system accessibility by allowing SK officials to submit reports, view announcements, access calendars, and receive notifications using mobile devices.

MySQL (Version 8.0, 2025)

MySQL is the relational database management system used to store and manage system data, including user records, barangay information, accomplishment reports, budget documents, chat logs, rankings, and analytics data. MySQL was used in SK 360° to ensure centralized, secure, and efficient data storage that supports synchronized access across both web and mobile platforms.

Firebase Cloud Messaging (Version 2025)

Firebase Cloud Messaging (FCM) is used to deliver real-time notifications such as announcements, submission reminders, and system updates. It helps SK 360° improve communication efficiency by ensuring that SK officials receive timely alerts and important updates directly through the system.

Dyte.io (Version 2025)

Dyte.io is used to provide real-time communication features such as chat, voice calls, and video conferencing within the SK 360° system. It enables virtual meetings, coordination sessions, and discussions among SK officials without relying on external communication platforms. Dyte.io was selected for SK 360° because of its reliable API, scalable infrastructure, and ease of integration with web and mobile applications. This technology enhances collaboration, supports remote governance activities, and improves communication efficiency across barangay and federation levels.

Bootstrap (Version 5.3, 2025)

Bootstrap 5.3 provides a responsive front-end framework that simplifies interface development. It ensures consistent design, mobile responsiveness, and usability across different screen sizes and devices. Bootstrap was used in SK 360° to speed up interface development while maintaining a clean and professional appearance.

Chart.js (Version 4.x, 2025)

Chart.js is used to generate graphical representations of analytics data such as report submission trends and engagement metrics. It helps SK 360° support monitoring and

evaluation by presenting data in an easy-to-understand visual format for federation-level reporting.

PHPMailer / Mailgun API (Version 2025)

PHPMailer and Mailgun API are used to send automated system emails such as account verification, notifications, and report-related alerts. These tools help SK 360° maintain reliable and secure communication with users regarding important system activities and updates.

XAMPP (Version 8.2.4, 2025)

XAMPP is utilized as a local development server environment that integrates Apache, PHP, and MySQL. It was used during the development of SK 360° to allow offline testing and debugging of backend functionalities before deployment, ensuring system stability and functionality.

Hostinger (Version 2025)

Hostinger serves as the web hosting platform for live deployment of the system. It provides secure hosting services, SSL encryption, and database support for continuous system availability. Hostinger helps SK 360° remain accessible online while protecting system data.

GitHub (Version 2025)

GitHub is used for version control and collaborative development. It allows the development team to manage source code, track changes, and maintain project documentation efficiently. GitHub helps ensure organized development and proper version tracking throughout the SK 360° project.

Table 3 Hardware Requirements for Systems Development

Hardware Component	Technical Specifications
Developer Workstation (Laptop/Desktop)	Desktop or laptop computer equipped with Intel Core i5 / AMD Ryzen 5 processor, 8 GB RAM, 256 GB SSD storage, and Windows 10 / Windows 11 or macOS 12 operating system.
Mobile Device (Testing Unit)	Android smartphone (version 10 or later) for testing the mobile application features

	such as notifications, chat, and report submission.
Server	Cloud-based hosting server with at least 2 CPU cores, 4 GB RAM, 50 GB SSD storage, and SSL certificate integration for secure data transmission.
Networking Devices	Stable internet connection with a minimum speed of 5 Mbps, Wi-Fi router, and Ethernet connectivity to support real-time communication and data synchronization.

Table 3 indicates that these are the hardware specifications for developing and testing the proposed system

Developer Workstation (Laptop/Desktop)

The developer workstation is used for system design, coding, debugging, and documentation. It runs the development tools, programming environments, and local servers needed to build and test the web and mobile applications efficiently.

Mobile Device (Testing Unit)

Mobile devices are used to test the mobile version of the system, ensuring proper functionality, responsiveness, and compatibility of features such as notifications, real-time communication, and report submission across supported Android platforms.

Server

The server hosts the deployed web application and database, managing user requests, data processing, and secure storage. It ensures system availability, scalability, and secure access for authorized users of the platform.

Networking Devices

Networking devices provide internet connectivity required for system development, testing, and deployment. A stable network connection supports real-time data synchronization, document uploads, and communication features of the system.

Network Requirements

The SK 360° system relies on a stable and secure internet connection to enable real-time data synchronization between the web application, mobile application, server, and database. The network configuration supports smooth API communication, fast system response, and continuous access to core features such as user authentication, accomplishment report submission, document uploads, announcements, real-time chat, video call/video chat, analytics dashboards, and notification services. Reliable network infrastructure is essential to ensure system availability, data consistency, and secure access for SK officials across barangays in Lipa City, especially since the platform supports real-time coordination and centralized documentation.

Wi-Fi (Wireless Fidelity) / Wired Connection

During system development, testing, and deployment, a wired or wireless internet connection is required for hosting, debugging, and synchronizing system components. A minimum internet speed of 5–10 Mbps is recommended to ensure efficient performance of both the web and mobile applications. This network setup enables seamless communication between the React.js web front-end, Laravel backend, and MySQL database running on a local development server (XAMPP) or the live hosting environment.

Mobile Data

When Wi-Fi or wired internet is unavailable, a mobile data connection with a minimum speed of 5 Mbps can be used to access and test system features. This ensures that SK officials can continue to submit reports, view announcements, participate in real-time chat, and receive notifications while on the go. The mobile application is optimized to handle varying mobile data speeds commonly available in the Philippines.

Cloud Hosting Server

The production environment of the SK 360° system is deployed on a cloud hosting platform (e.g., Hostinger, 2025) to ensure scalability, security, and continuous availability. The cloud server supports SSL encryption for secure data transmission and enables centralized data storage and synchronization between web and mobile platforms. It manages real-time updates for announcements, notifications, chat messages, and

analytics data, ensuring reliable system access for all authorized users within the SK Federation of Lipa City.

Peopleware Requirements

Project Manager

The project manager oversees the entire development lifecycle of SK 360°, coordinates team activities, manages timelines, resources, and budgets, and ensures alignment with project objectives and academic standards. This role facilitates communication among team members, the capstone adviser, and the client (SK Federation of Lipa City), monitors progress, mitigates risks, and delivers regular status reports to maintain project momentum and successful completion within the research timeframe.

System Analyst

The system analyst is responsible for gathering and analyzing system requirements from the SK Federation of Lipa City. This role ensures that user needs are properly translated into system features, workflows, and technical specifications before development begins.

UI/UX Designer

The UI/UX designer is responsible for designing the system's interface and user experience. This includes creating layouts, navigation flows, and visual elements that ensure the system is user-friendly, accessible, and suitable for SK officials and youth users.

Front-End Developer

The front-end developer builds the user interface of the web and mobile applications using React.js and Flutter. This role ensures that all visual components function correctly and that the system is responsive across different devices.

Back-End Developer

The back-end developer handles server-side logic using PHP and Laravel. This includes implementing authentication, data processing, system rules, and API integration between the client interface and the database.

Database Administrator

The database administrator is responsible for designing, managing, and maintaining the MySQL database. This role ensures data integrity, security, backup, and efficient retrieval of records such as reports, documents, user accounts, and announcements.

Quality Assurance (QA) Tester

The QA tester performs system testing to identify bugs, errors, and performance issues. This role ensures that all system features function as intended and that the platform meets usability and reliability standards before deployment.

Capstone Adviser

The project adviser provides technical guidance, monitors progress, and ensures that the development follows academic standards and project requirements. The adviser also evaluates system quality and provides recommendations for improvement.

Table 4 Project Team Members, Roles and Contact Details

Name(s)	Role	Contact Details
CAPSTONE PROJECT TEAM		
Bautista, Marielle O.	Back End Developer/UI/UX Designer	bautistamo@students.nu-lipa.edu.ph
Janda, Giorgia Schen C.	Database Administrator/System Analyst	jandagc@students.nu-lipa.edu.ph
Monje, Paul Vincent M.	Back End Developer/QA Tester	monjepm@students.nu-lipa.edu.ph
Sumadsad, Alexa E.	Project Manager/UI/UX Designer/ Front-End Developer	sumadsadae@students.nu-lipa.edu.ph
Natividad, Reymark R.	Capstone Adviser	
CLIENT/PARTNER ORGANIZATION TEAM REPRESENTATIVES		
Sangguniang Kabataan Federation of Lipa City		

Table 4 indicates the individuals involved in developing and testing the proposed system.

3.2 PROJECT TECHNICAL DESCRIPTION

The proposed project, SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City, is a dual-platform system designed to operate on both web and mobile environments. It provides a centralized digital platform where the SK Federation and barangay SK councils can manage accomplishment reports, budget documentation, announcements, schedules, and communication in a secure and organized manner. The system supports role-based access for SK officials while allowing Kabataan limited access to public information such as announcements, event calendars, and leadership profiles. By integrating reporting, coordination, and communication features into a single platform, the system aims to improve efficiency, transparency, and continuity in youth governance processes. Additionally, the system is designed to reduce manual workload and administrative delays commonly experienced in SK operations by automating submissions, notifications, and data consolidation.

The development of SK 360° utilizes modern web and mobile technologies, including PHP and Laravel for backend processing and system logic, MySQL for centralized data storage, and React.js and Flutter for the web and mobile user interfaces, respectively. The system follows a three-tier client–server architecture, where the presentation layer handles user interaction, the application layer processes system requests and enforces business rules, and the data layer manages secure data storage and retrieval. This architectural design ensures organized system flow, efficient data management, and real-time synchronization of information such as report submissions, announcements, chat messages, and notifications across web and mobile platforms. The modular structure of the system also allows individual components to be enhanced or maintained independently without affecting the overall system performance.

As illustrated in Figure 1, the overall system architecture of SK 360°, illustrating a three-tier client-server model that separates concerns for better maintainability, scalability, and security. The presentation layer consists of the user interfaces: the React.js-based web application (accessible via browsers on desktops/laptops) and the Flutter-based mobile application (for Android and iOS devices). These clients handle user interactions, input validation, and visual rendering, ensuring responsive and intuitive experiences across devices. Users such as SK Federation President/Admin, SK Chairman, SK Secretary, barangay SK officials, and limited public viewers (kabataan) interact through these

interfaces to access features like report submission, announcements, real-time chat, and analytics dashboards. This layer also ensures accessibility for users with varying levels of technical expertise through simplified navigation and consistent interface design.

The application layer (backend) is powered by PHP with Laravel, serving as the core business logic engine. It processes requests from the clients via RESTful APIs, enforces role-based access control (RBAC), handles authentication (e.g., JWT tokens or session management), and orchestrates operations such as report consolidation, document uploads, notification triggers via Firebase Cloud Messaging, real-time communication via Dyte.io, and gamified ranking calculations. Business rules are strictly enforced at this layer to ensure data integrity, security, and compliance with defined SK governance workflows. This layer ensures secure data flow, business rules (e.g., restricting sensitive uploads to authorized roles), and real-time synchronization between web and mobile platforms.

The data layer utilizes MySQL as the relational database, storing structured data including user profiles, accomplishment reports, budget documents, chat logs, leadership records, announcements, and analytics metrics. Data is persisted securely with encryption at rest and in transit (via SSL/TLS on Hostinger hosting), while backups and logging support system reliability and auditability. Versioned backups and historical data retention further support leadership transitions and long-term record preservation.

This architecture promotes scalability by allowing independent scaling of layers (e.g., adding more server instances for high traffic during reporting periods) and security through layered isolation (e.g., preventing direct client-database access). It aligns perfectly with SK 360°'s requirements for centralized coordination, real-time features, and role-specific access in Lipa City's 72-barangay context, enabling efficient, transparent youth governance without compromising performance or data integrity. Overall, the technical design ensures that the system remains adaptable to future enhancements and increasing user demands.

Special Features of the System

SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City introduces an integrated digital solution that addresses the common challenges faced by the SK

Federation and barangay SK councils in managing reports, budgets, coordination, and communication. Unlike traditional paper-based and fragmented digital methods, the system consolidates governance processes into a single, secure, and accessible platform. By leveraging modern web and mobile technologies, SK 360° promotes transparency, efficiency, and continuity in youth governance operations while supporting real-time collaboration among SK officials and stakeholders. The system is designed to support both administrative efficiency and public transparency by balancing secure internal operations with accessible public information. Furthermore, the platform standardizes governance workflows, enabling consistent implementation of policies, reporting procedures, and communication practices across all barangay SK councils. This integrated approach helps strengthen coordination between the SK Federation and barangay levels while ensuring reliable access to accurate and up-to-date information.

Centralized Accomplishment and Budget Documentation

The system provides a centralized repository where barangay SK councils can upload, manage, and track accomplishment reports, financial documents, and supporting files. This feature minimizes document loss, reduces redundancy, and ensures that all records are properly archived and accessible to authorized users, particularly the SK Federation and administrators. Standardized submission formats and centralized storage also simplify validation, review, and long-term record keeping.

Role-Based Access Control

SK 360° implements role-based access to ensure data security and accountability. Different user roles such as SK Federation administrators, barangay SK officials, and public viewers are granted specific permissions based on their responsibilities. This prevents unauthorized access to sensitive documents while allowing transparency for public-facing information such as announcements and programs. Role-based controls further strengthen accountability by clearly defining user responsibilities within the system.

Real-Time Notifications and Announcements

The platform integrates real-time notifications using Firebase Cloud Messaging to keep users informed of important updates such as report deadlines, approval status, announcements, and upcoming activities. This feature improves coordination and reduces

delays in communication between the SK Federation and barangay SK councils. Automated alerts help ensure timely compliance with reporting and activity requirements.

Integrated Communication and Messaging System

SK 360° includes a built-in messaging feature that allows SK officials to communicate directly within the platform. This eliminates the need to rely on external messaging applications, ensuring that all official discussions, clarifications, and coordination remain documented and centralized. Maintaining communication records within the system also supports transparency and accountability in official interactions.

Calendar and Event Management

The system includes a digital calendar for scheduling SK activities, meetings, deadlines, and youth programs. This feature helps users monitor upcoming events and avoid scheduling conflicts, promoting better time management and organized program implementation. Shared visibility of schedules improves coordination across barangay and federation levels.

Gamified Ranking and Performance Points

The system incorporates a gamified ranking and performance points mechanism to encourage active participation and timely submission of reports among barangay SK councils. Points are awarded based on compliance, engagement, and completion of required tasks, allowing the SK Federation to monitor performance in a transparent and motivating manner while fostering healthy competition among SK officials. This feature promotes sustained engagement and recognizes consistent performance among participating councils.

Leadership Transition Support and Centralized Archiving

SK 360° includes a centralized document archive designed to support smooth leadership transitions within the SK councils. This feature securely preserves historical records such as reports, budgets, and official documents, ensuring continuity of governance and preventing data loss when leadership roles change across different SK terms. Archived records remain accessible to authorized users for reference, evaluation, and auditing purposes.

Analytics and Monitoring Dashboard

An analytics dashboard presents summarized data such as submitted reports, pending documents, completed programs, and budget utilization. This allows the SK Federation to monitor the performance of barangay SK councils and supports data-driven decision-making and evaluation. Visual summaries and indicators enable faster assessment of compliance and overall performance.

Cloud Hosting and Data Security

The platform is deployed on a cloud-based hosting server with SSL encryption to ensure secure data transmission and storage. Sensitive information such as budget files, reports, and user credentials is protected through authentication mechanisms and database security controls, ensuring system reliability and data integrity. Regular backups and access monitoring further enhance system security and reliability.

Cross-Platform Accessibility

Developed using React.js for the web and Flutter for mobile, SK 360° provides a consistent and responsive user experience across desktop and mobile devices. This allows SK officials to access the system anytime and anywhere, improving flexibility and responsiveness in governance activities. Cross-platform support ensures wider system adoption and uninterrupted access across devices.

User-Centered Interface Design

The system follows a clean and intuitive UI/UX design using Bootstrap 5.3. Simple navigation, readable layouts, and clear icons ensure that users with varying levels of technical skills can easily use the platform, reducing the learning curve and increasing user adoption. This design approach enhances usability and encourages consistent system usage among all user groups.

System Architecture

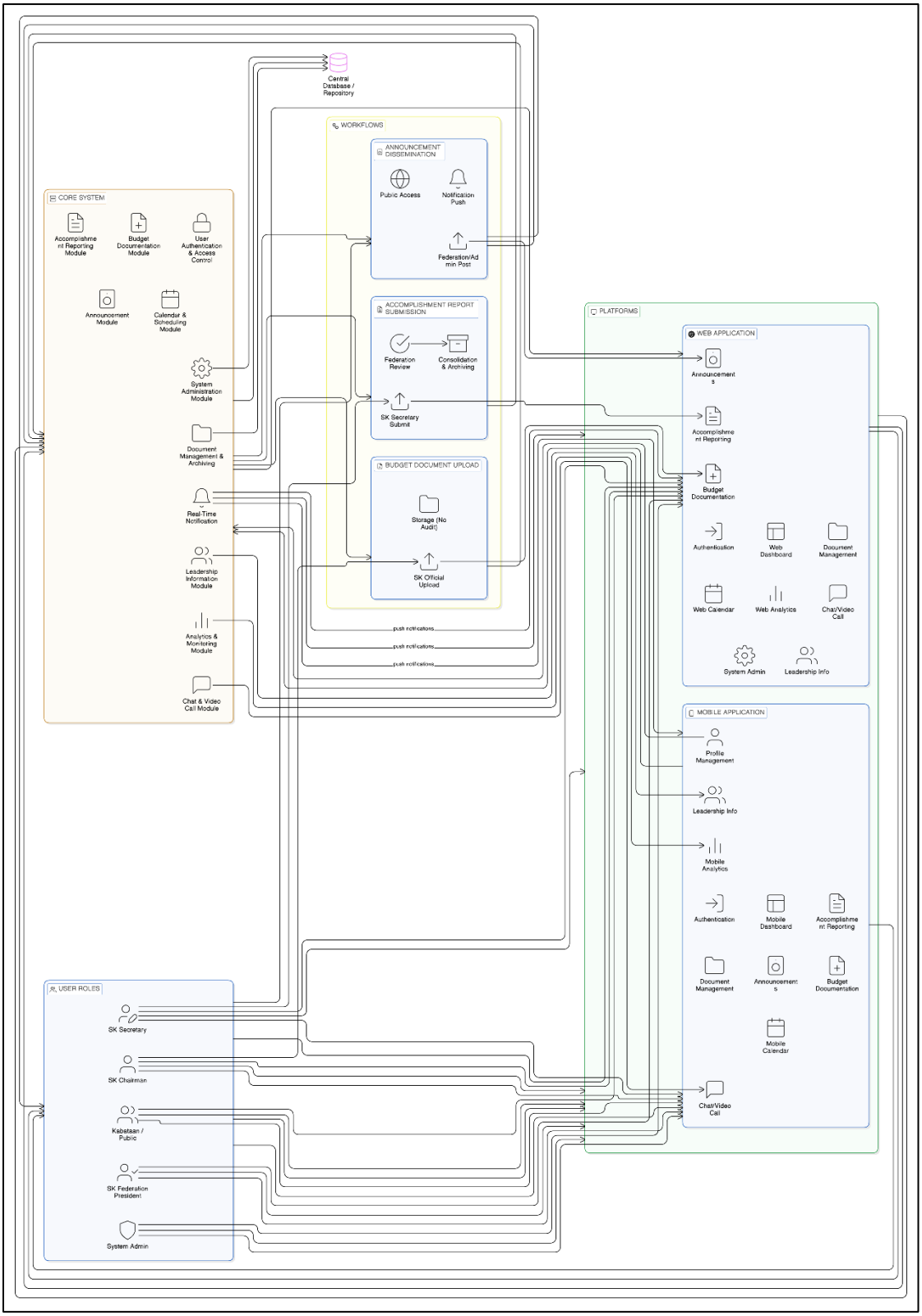
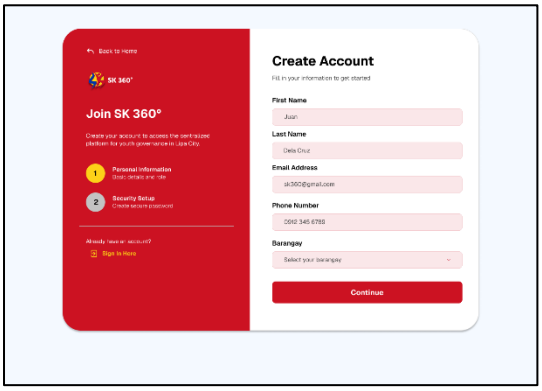


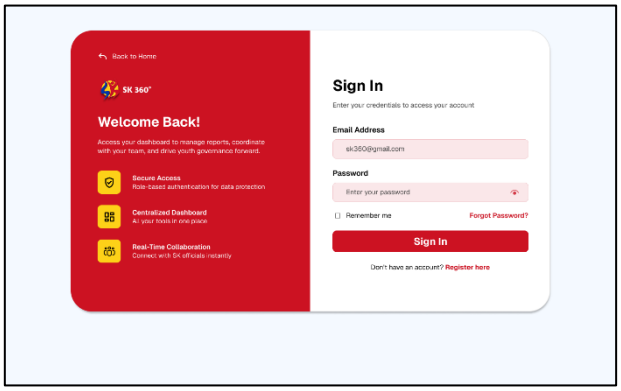
Figure 1 Overall System Architecture

Web Application Modules

User Authentication and Access Control Module

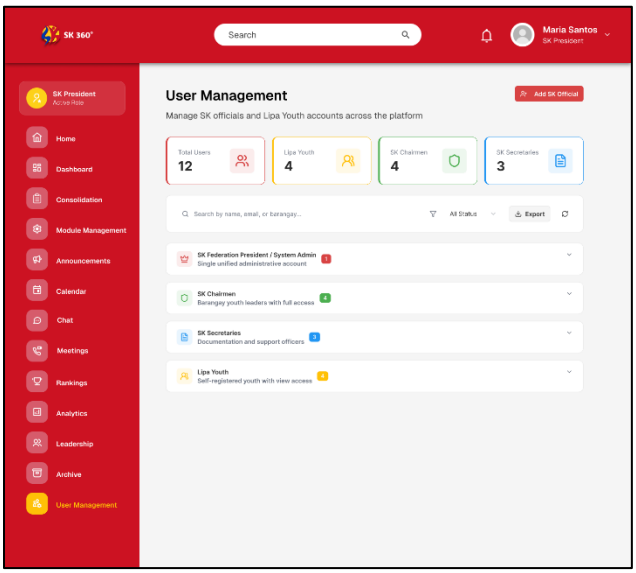
This module manages user registration, login, and access permissions within the system. It ensures that only authorized users can access specific features based on their assigned roles, such as SK Federation President, barangay SK officials, and Lipa Youth. The module includes secure login credentials, session management, and password encryption to protect user accounts and system data.





User Management Module

The User Management Module allows the SK Preisdent to manage user accounts within the system. It supports creating, updating, activating, or deactivating accounts of SK officials. This module ensures that only verified and authorized individuals can submit reports, upload documents, and access internal system features.

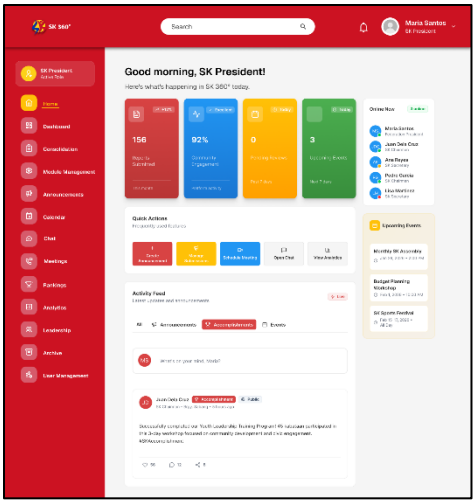


Dashboard Module

Serves as the main landing page after user login, providing a role-based overview of key system information such as pending and submitted reports, recent announcements, upcoming activities, notifications, and performance summaries. It presents visual insights using charts and widgets to help users quickly monitor status, priorities, and overall participation within the SK system.

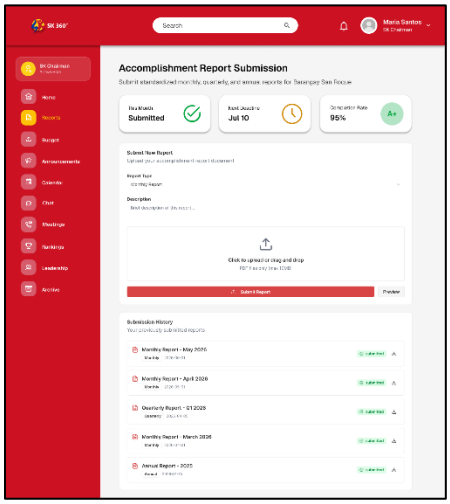
Home Page Module

Serves as the system's main entry point, providing users with an overview of the platform, including system highlights, key features, public announcements, upcoming activities, and quick navigation links. For logged-in users, the Home Page adapts to display role-based summaries and directs them efficiently to relevant modules, ensuring ease of use, accessibility, and a clear starting point for all users.



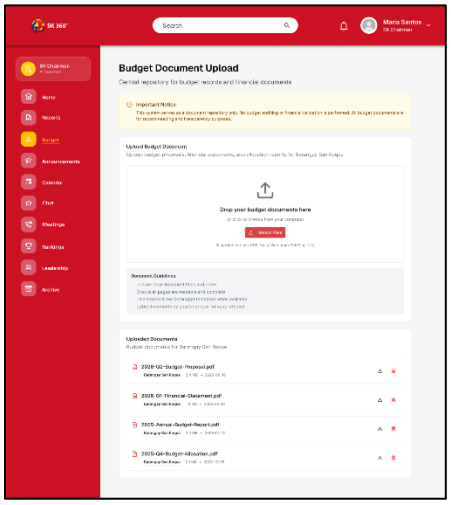
Accomplishment Reporting Module

This module enables barangay SK councils to submit digital accomplishment reports in a standardized format. Users can input activity details, attach supporting documents or photos, and submit reports directly to the SK Federation. The system automatically organizes submissions by barangay and reporting period, reducing manual consolidation and paperwork.



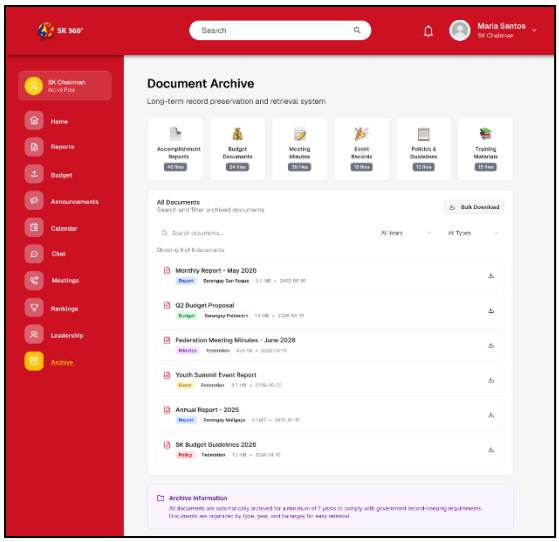
Budget and Financial Documentation Module

The Budget and Financial Documentation Module handles the submission and storage of budget proposals, liquidation reports, and financial records. It allows SK officials to upload documents securely while enabling the SK Federation to review, monitor, and verify submitted financial data. This module promotes transparency and accountability in SK financial management.



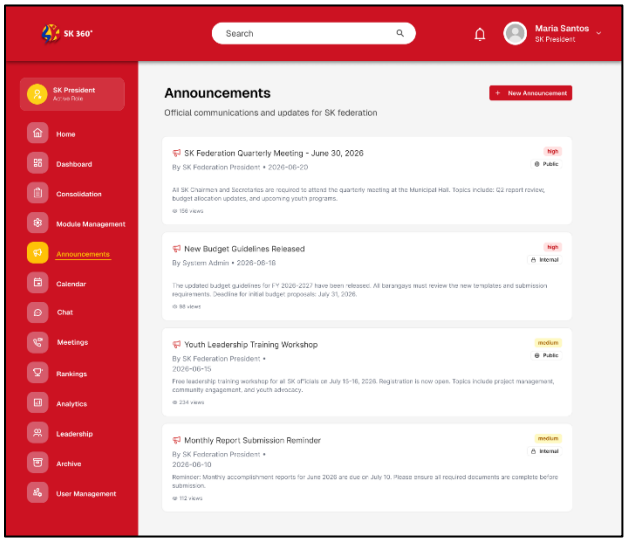
Document Management and Archiving Module

This module provides centralized storage for official SK documents such as resolutions, ordinances, memoranda, and supporting attachments. It includes file categorization, search functionality, and download options, ensuring that documents are easy to retrieve and protected from unauthorized access.



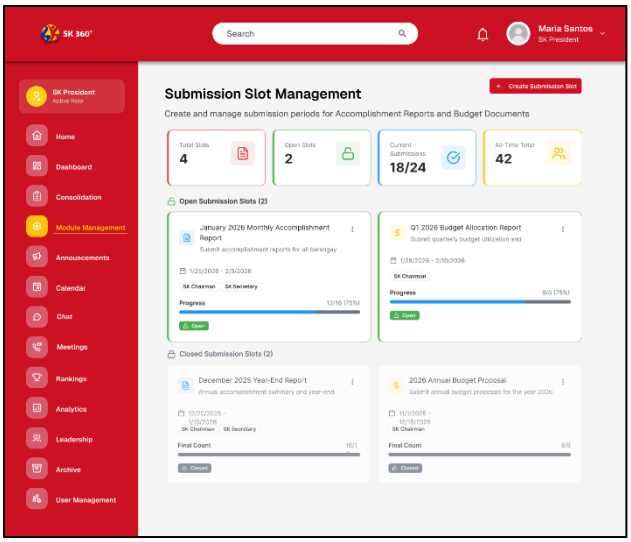
Announcement and Information Module

The Announcement Module allows the SK Federation to post official announcements, memos, and updates visible to barangay SK officials and, when applicable, the public. This ensures timely dissemination of information such as meeting schedules, deadlines, and events, improving coordination across all SK units.



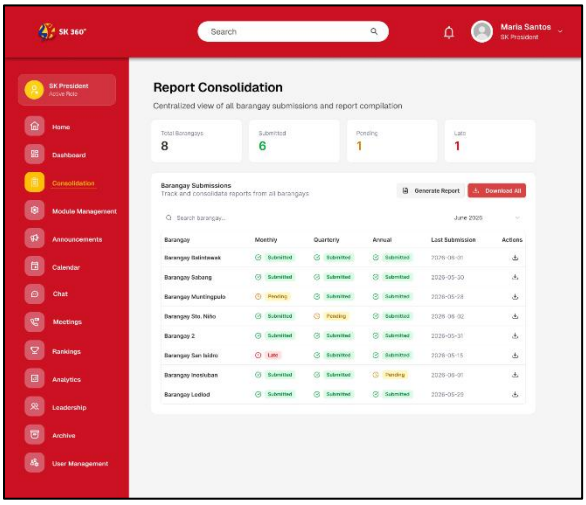
Submission Management Module

This module allows the SK Federation President or authorized administrator to create, configure, and manage submission slots for specific SK roles such as the SK Chairman and SK Secretary. It enables the setting of submission periods, deadlines, and role-based access to reporting and documentation modules. The system ensures that only designated officials can submit required reports within the approved timeframe, improving submission control, accountability, and workflow efficiency.



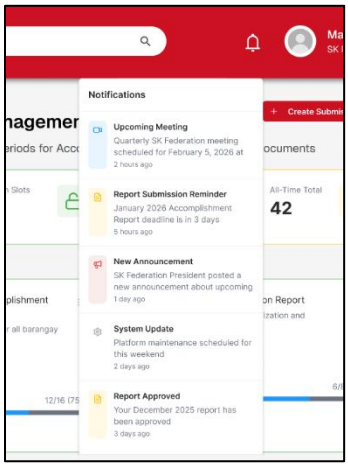
Consolidation Reports Module

Automates the compilation of monthly, quarterly, and annual accomplishment reports from barangay submissions into unified PDF formats, allowing the SK Federation to review, consolidate, and archive citywide data for oversight and reporting to DILG/COA.



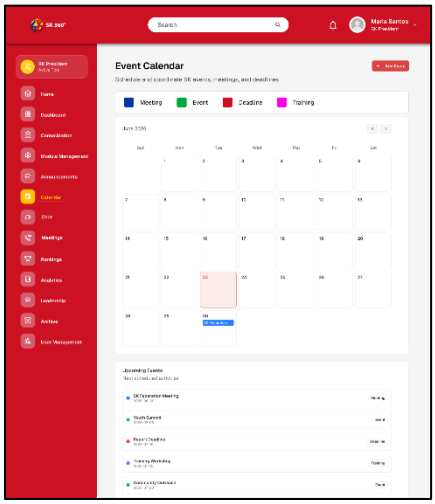
Notification Module

This module delivers real-time system notifications to users regarding report submissions, approval status, deadlines, and announcements. Notifications help users stay informed and reduce missed tasks by providing alerts through the web interface and synchronized mobile notifications.



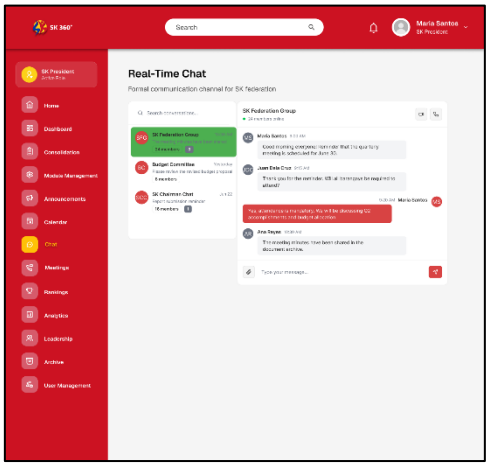
Activity Calendar and Scheduling Module

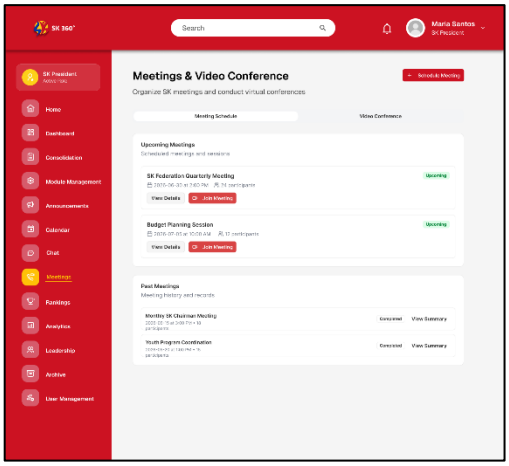
The Activity Calendar Module displays scheduled SK activities, meetings, and events in a centralized calendar view. Users can view upcoming activities, deadlines, and important dates, helping improve planning and time management among SK officials.



Chat and Meetings / Video Conference Module

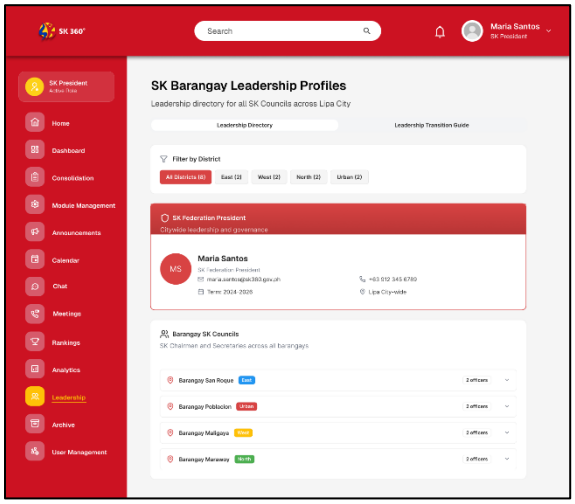
Enables real-time communication among SK officials through integrated chat, voice calls, and video conferencing using Dyte.io. This module supports virtual meetings, coordination, and collaboration across barangays, with discussions and meeting sessions documented within the system to ensure accessibility, accountability, and effective remote communication.





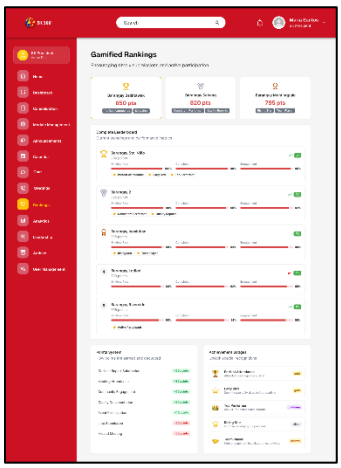
Leadership Information Module

Provides a centralized directory of current SK Federation and barangay SK officials, displaying verified details such as names, positions, barangays, and term coverage. It promotes transparency by allowing users, including the Kabataan, to easily identify youth leaders and supports administrators in updating leadership records when changes occur.



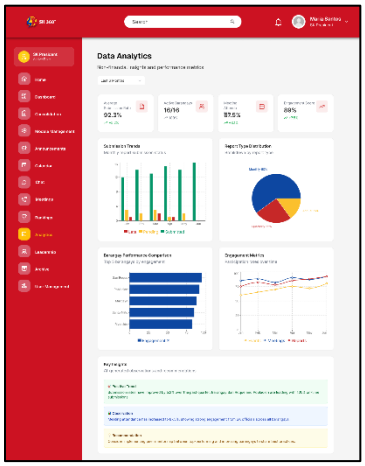
Gamified Ranking and Incentive Module

This module introduces gamification elements to encourage active participation and timely compliance among barangay SK officials. It generates performance-based rankings based on criteria such as report submission timeliness, completeness, and activity participation. The module displays leaderboards for barangays and SK officials, allowing the SK Federation to recognize high-performing councils and motivate continuous engagement. Gamification features support transparency, accountability, and increased user participation in system activities.



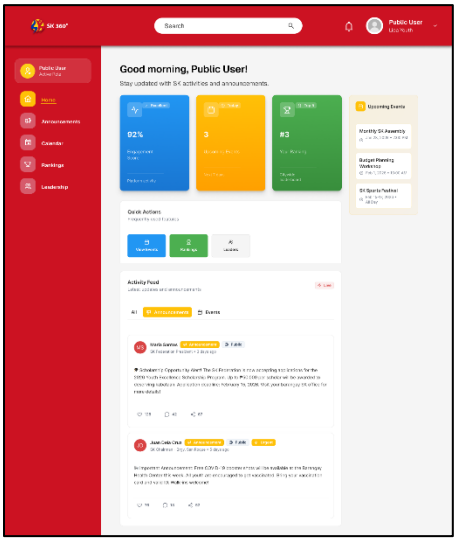
Analytics and Monitoring Module

This module provides summarized data and visual reports using charts and graphs. It allows the SK Federation to monitor submission status, barangay compliance, and activity distribution. The analytics support decision-making and help identify gaps or delays in reporting.



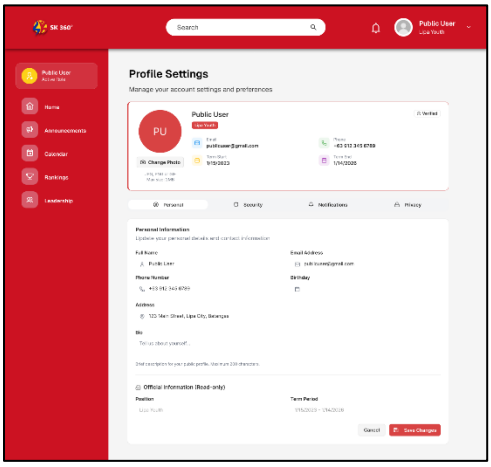
Public View Module

Provides limited, read-only access to the general Kabataan of Lipa City, allowing them to view publicly available content such as official announcements, event calendars, leadership information, and gamified rankings. This module promotes transparency, youth engagement, and public awareness without exposing sensitive system data or internal functionalities.



Profile Settings Module

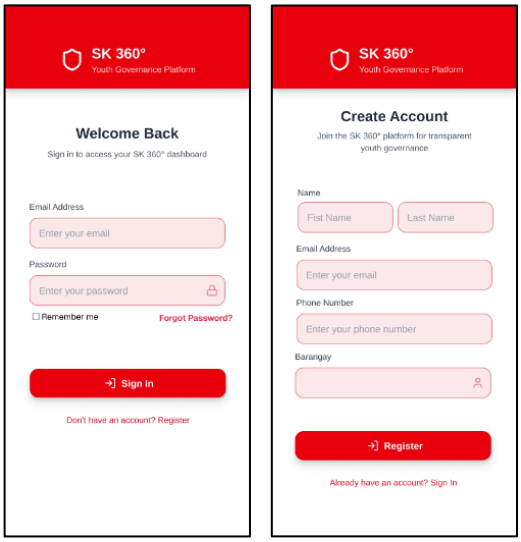
Allows users to manage and update their personal account information, including profile details, contact information, passwords, and notification preferences. This module ensures secure account management while complying with data privacy standards (e.g., RA 10173) and maintaining the integrity of verified leadership records.



Mobile Application Modules

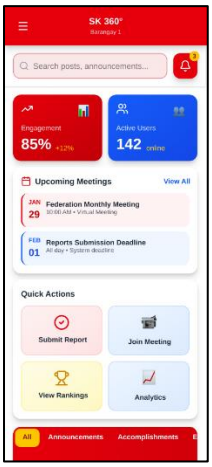
Mobile User Authentication Module

This module manages secure login and session handling for mobile users. It allows SK officials and authorized users to access their accounts using encrypted credentials while ensuring role-based access control. The module maintains secure sessions and protects user data even when accessed through mobile networks.



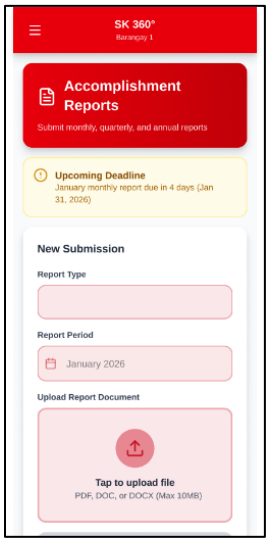
Home Page Module

The Home Page Module serves as the main entry point of the mobile application. It provides users with a quick overview of important information such as recent announcements, upcoming activities, rankings highlights, and shortcuts to accessible modules based on user role, ensuring easy navigation and immediate access to key features.



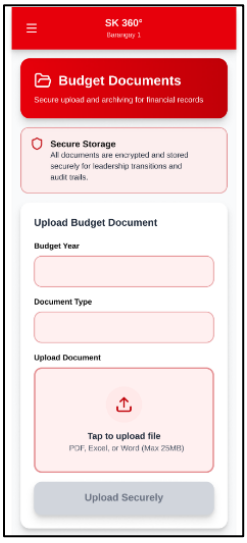
Accomplishment Reporting Module (Mobile)

This module enables SK officials to submit accomplishment reports directly through their mobile devices. Users can enter activity details, upload photos or documents, and submit reports in real time. This feature supports on-site reporting, especially during or immediately after SK activities.



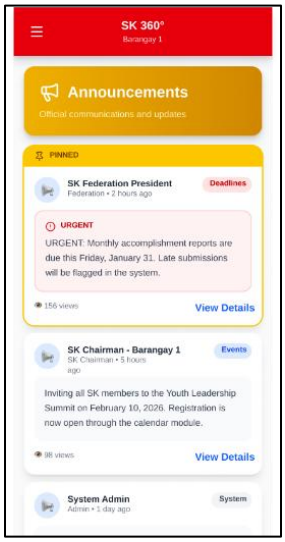
Budget Documents Module

The Budget Documents Module enables authorized users to upload, view, and access budget-related files such as proposals and liquidation documents through the mobile application. Uploaded files are securely stored in the centralized database to support transparency, record-keeping, and easy retrieval.



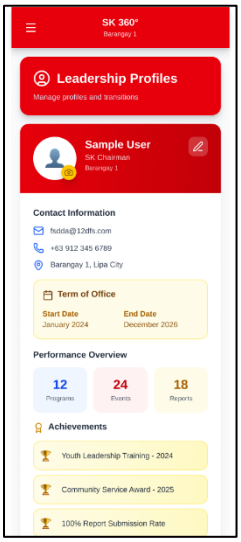
Announcement Module

This module displays official announcements, memos, and updates issued by the SK Federation. It ensures that users are informed of important notices, deadlines, and events through a centralized and mobile-accessible announcement feed.



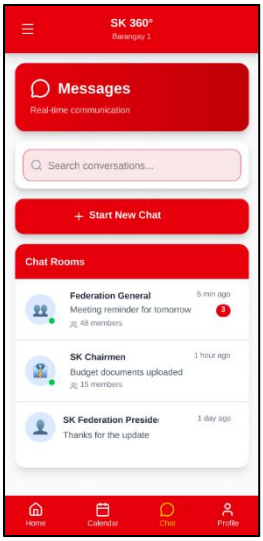
Leadership Profiles Module

The Leadership Profiles Module provides mobile access to verified information about SK Federation and barangay SK officials. It displays names, positions, assigned barangays, and term coverage, promoting transparency and allowing users and Kabataan to easily identify youth leaders.



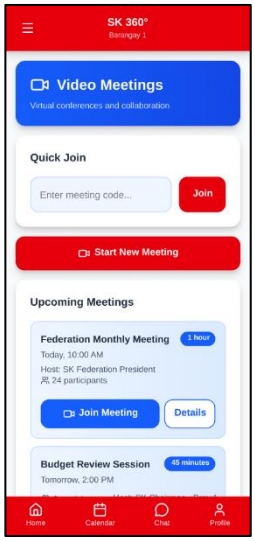
Chat Module

The Chat Module enables real-time messaging among SK officials using mobile devices. It supports efficient coordination, quick clarifications, and continuous communication without relying on external messaging platforms.



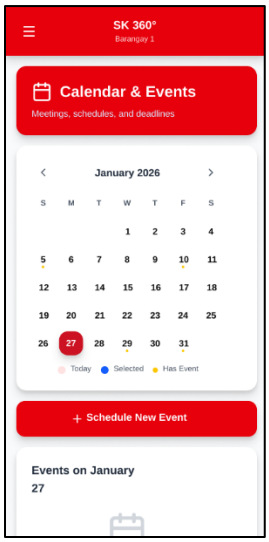
Video Meetings Module

This module allows users to participate in virtual meetings through integrated video conferencing on mobile devices. It supports coordination and collaboration across barangays, reducing the need for physical meetings and overcoming logistical constraints.



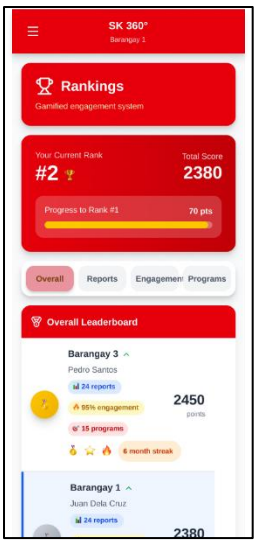
Activity Calendar Module

The Activity Calendar Module displays scheduled SK activities, meetings, and events in a mobile-friendly calendar view. It helps users track important dates, avoid scheduling conflicts, and stay updated on upcoming SK engagements.



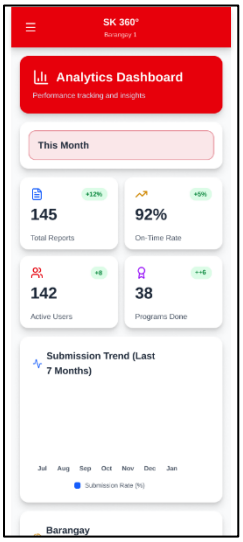
Rankings Module

The Rankings Module presents gamified performance rankings of barangay SK councils through the mobile application. Leaderboards and compliance indicators motivate participation, timely submissions, and active engagement in SK activities.



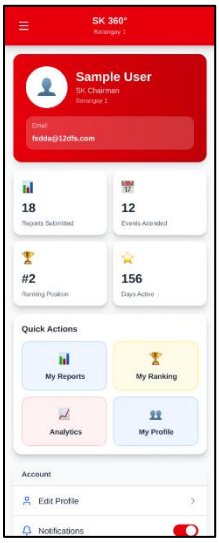
Analytics Module

This module provides mobile access to summarized analytics such as submission status and participation trends. It supports monitoring and decision-making by presenting key insights in a simplified and easy-to-understand format.



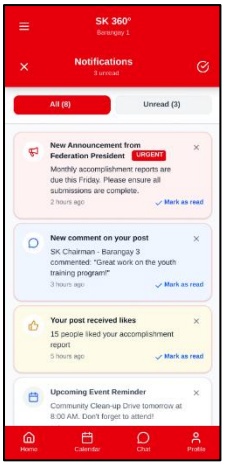
Profile Module

The Profile Module allows users to manage their personal account information, including contact details and passwords, using their mobile devices. It supports secure account maintenance while complying with data privacy requirements.



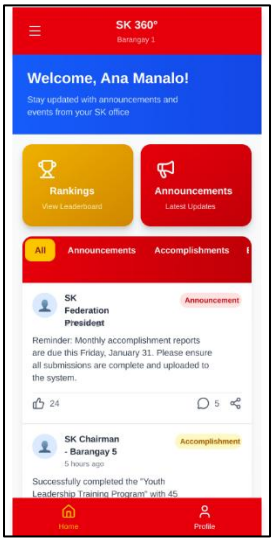
Notification Module

The Notification Module delivers real-time alerts to users through the mobile application regarding announcements, report submissions, deadlines, approvals, meetings, and system updates. By using push notifications, this module helps reduce missed tasks, improves responsiveness, and supports timely coordination among SK officials while keeping Kabataan informed of public announcements.



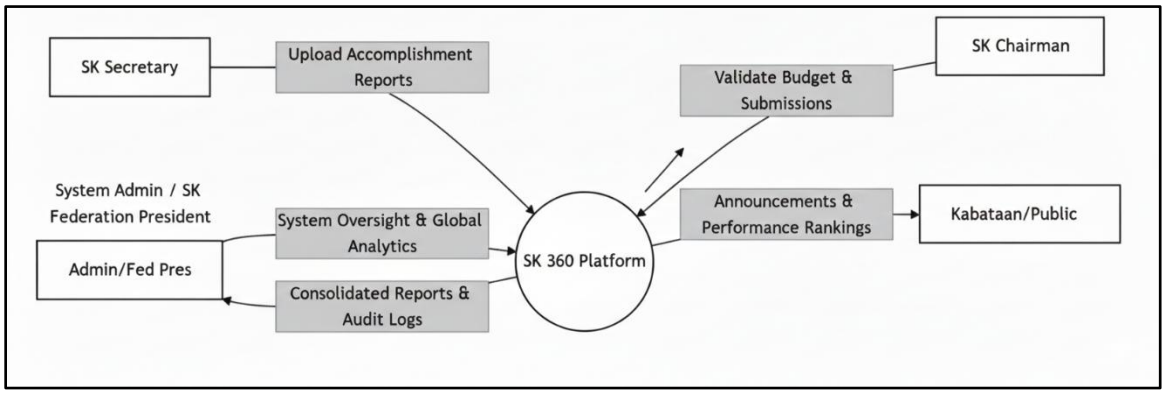
Public View Module

The Public View Module provides limited, read-only access to non-registered users and Kabataan through the mobile application. It allows viewing of public information such as official announcements, activity calendars, rankings, and leadership profiles, promoting transparency and youth awareness while ensuring that sensitive system features and data remain restricted to authorized SK officials.

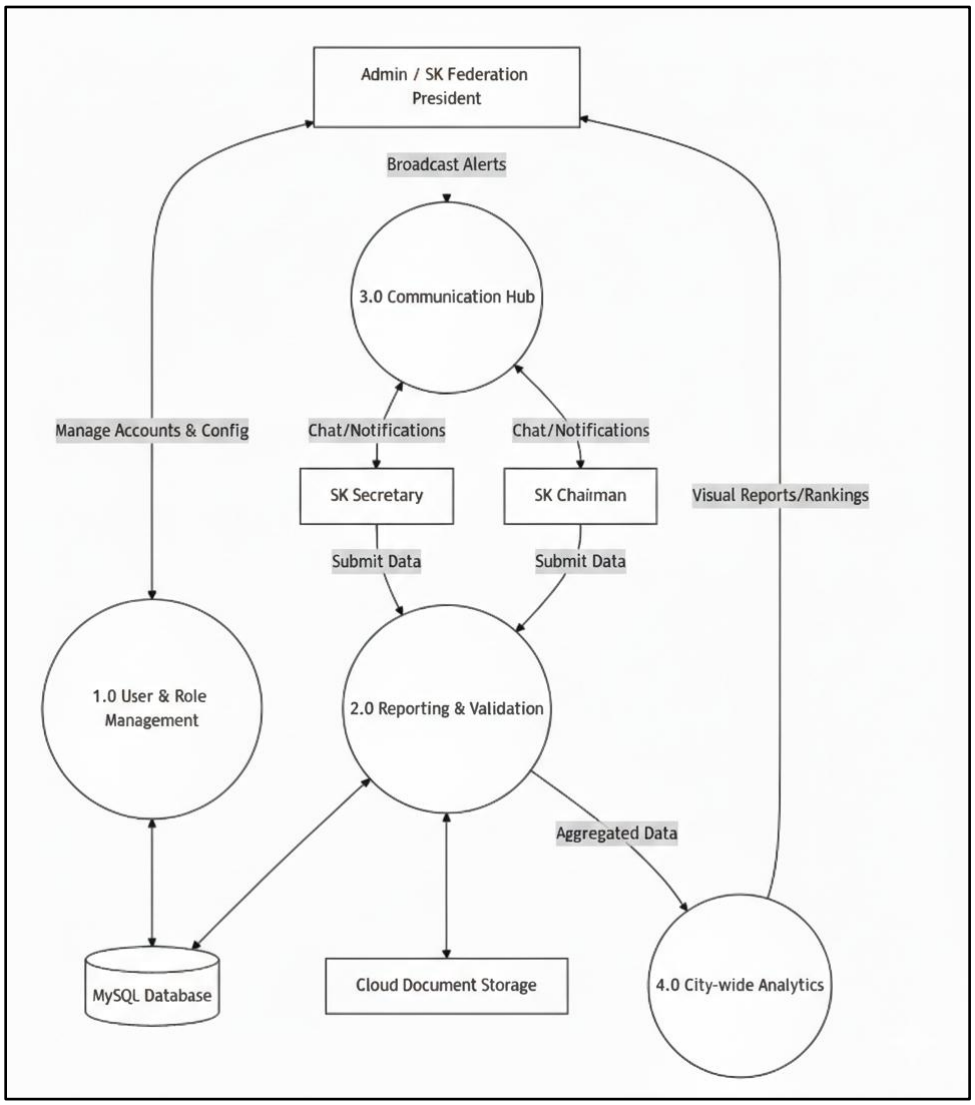


Data Flow Diagram

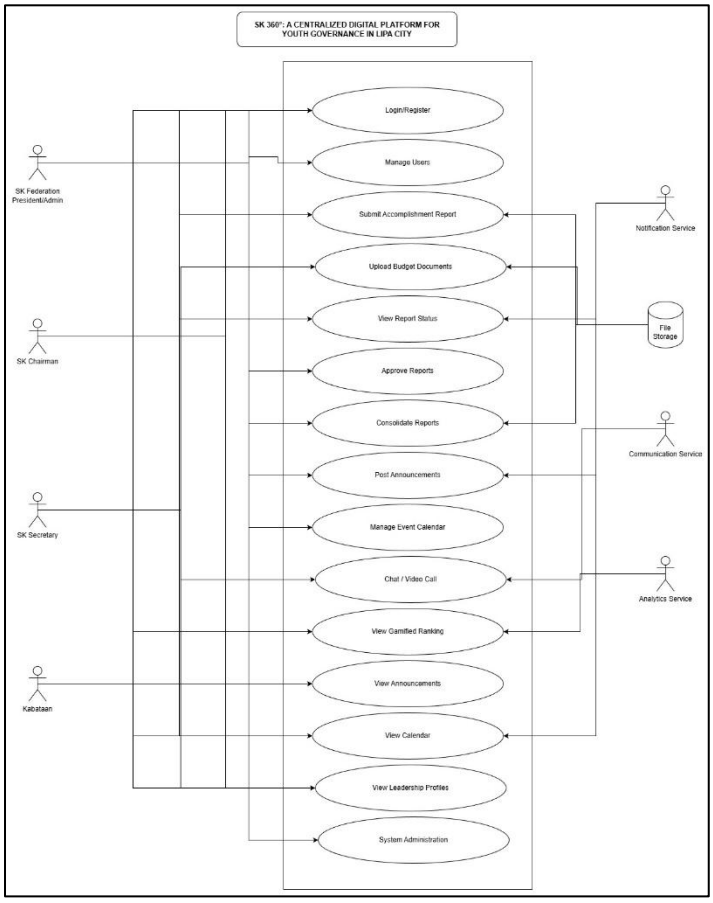
Level 0



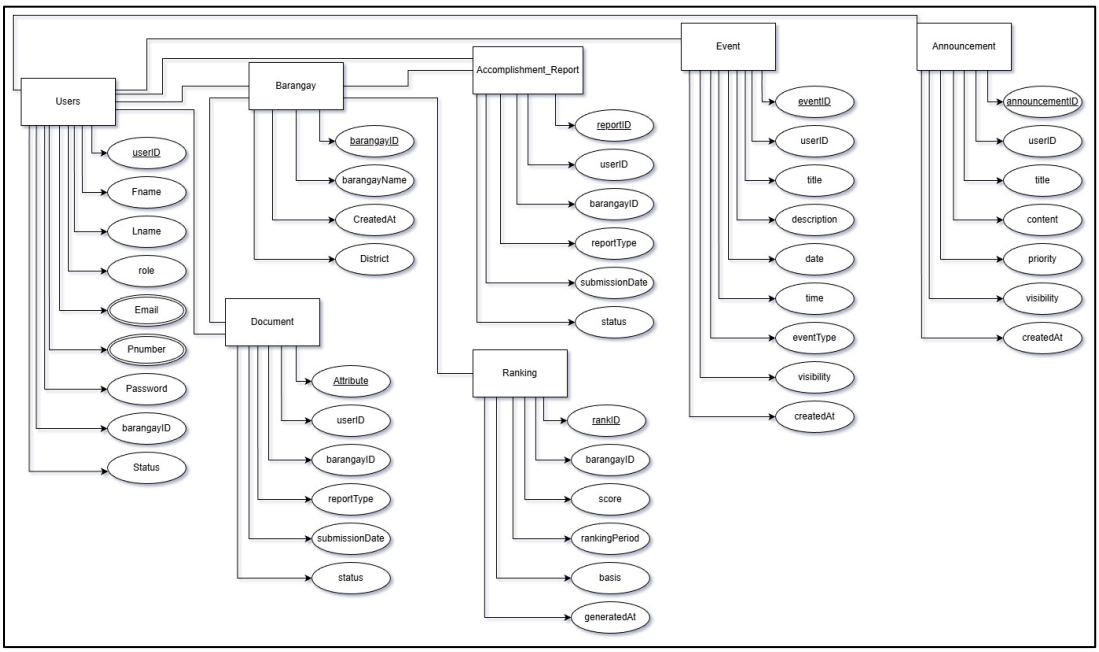
Level 1



Use Case Diagram - Unified Modeling Language (UML)



Entity Relationship Diagram



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CHAPTER 4
METHODOLOGY, SYSTEM PROCESSES AND DESIGNS

This chapter presents the methodology, requirement analysis and documentation, the design of the system using different models and the development of the proposed system application.

4.1 REQUIREMENTS ANALYSIS AND DOCUMENTATION

This section of your capstone manuscript contains the employed methods to address the project proposal which comprises the following sub-sections:

Research Methodology Used

Discuss thoroughly the research methodology pursued by the proponents. A research methodology describes the techniques and procedures used to identify and analyze information regarding a specific research topic. It is a process by which researchers design their study so that they can achieve their objectives using the selected research instruments. It includes all the important aspects of research, including research design, data collection methods, data analysis methods, and the overall framework within which the research is conducted. There are three types of research methodology—quantitative, qualitative, and mixed-method, which can be chosen based on the research objectives.

Project Development Methodology

Select the appropriate project development methodology that fits on your proposed project such as Agile, Waterfall, etc. Include here your own illustration of the project development methodology framework used (MWA – Agile, Waterfall; MAA – Multimedia Cycle).

Figure 2 Project Development Methodology

Discuss the various stages in the project development methodology vis-à-vis on tasks/requirements of your proposed project. Also, in this section include a table that list the detailed activities on various phase/stages of the project development framework and the required deliverables.

Table 5 Project Development Phases, Activities and Deliverables

Phases	Activities	Deliverables

Functional Requirements

In this section of your manuscript, be able to provide a table that presents the list of system modules (MWA – both web and mobile system; MAA – either web, desktop or mobile system) based on required functionalities (user-defined, proponent-defined, and/or panelist-defined functions). Moreover, detailed functional requirement should be included in the Appendix section of your manuscript.

Table 6 Functional Requirements of the Project

Functional Requirement ID	Function Name(s)	Description

Non-Functional Requirements

In this section of your manuscript, be able to provide a table that presents the list of non-functional requirements (MWA – both web and mobile system; MAA – either web, desktop or mobile system) which are a set of specifications that describe the system's operation capabilities and constraints such as the following: Functionality, Performance, Scalability, Portability, Compatibility, Reliability, Availability, Maintainability, Security, and Usability.

Table 7 Non-Functional Requirements of the Project

Non-Functional Criteria	Description
	The system should: 1. 2. The system is constrained to: 1. 2.

4.2 DESIGN OF SOFTWARE, SYSTEMS, PRODUCTS AND/OR PROCESSES

Use Case Diagram

Figure 3 Use Case Diagram

Present and discuss in this paragraph the Use Case Diagram of the System (MWA – both web and mobile system; MAA – either web, desktop or mobile system) (Use Style: Capstone Content – Body Text). Note: Diagrams symbols, flows and labels must be readable.

Use Case Narratives

Present in this section the Use Case Narratives based on various Use Case (MWA – both web and mobile system; MAA – either web, desktop or mobile system) illustrated in the Use Case Diagram. Narratives should be consistent.

Table 8 Use Case Narratives of the Web Application

Use Case Name	
Actor(s)	
Description	
Pre-condition	
Post-condition	
Main Scenario	
Alternative Scenario	

Use Case Name	
Actor(s)	
Description	
Pre-condition	
Post-condition	
Main Scenario	
Alternative Scenario	

Use Case Name	
Actor(s)	
Description	
Pre-condition	
Post-condition	
Main Scenario	
Alternative Scenario	

Use Case Name	
Actor(s)	
Description	
Pre-condition	
Post-condition	
Main Scenario	
Alternative Scenario	

Table 9 Use Case Narratives of the Mobile Application

Use Case Name	
Actor(s)	
Description	
Pre-condition	
Post-condition	
Main Scenario	
Alternative Scenario	

Use Case Name	
Actor(s)	
Description	
Pre-condition	
Post-condition	
Main Scenario	
Alternative Scenario	

Functional Decomposition Diagrams

Figure 4 Functional Decomposition Diagram of the Web Application

Present and discuss in this paragraph the Functional Decomposition Diagram of the system (MWA – both web and mobile system; MAA – either web or mobile system, if applicable otherwise MAA required designs) (Use Style: Capstone Content – Body Text). Note: Diagrams symbols, flows and labels must be readable.

Figure 5 Functional Decomposition Diagram for Mobile Application

Present and discuss in this paragraph the Functional Decomposition Diagram of the system (MWA – both web and mobile system; MAA – either web or mobile system, if applicable otherwise MAA required designs) (Use Style: Capstone Content – Body Text). Note: Diagrams symbols, flows and labels must be readable.

Process Flow Diagrams

Figure 6 Process Flow – Module A

Present in this section all the process flow of the system (users vis-à-vis system modules) using Activity Diagram with Swim Lanes (MWA – both web and mobile system; MAA – either web or mobile system, if applicable otherwise MAA required designs). Describe each illustration explaining the various processes involved in the specific module (Use Style: Capstone Content – Body Text). Note: Diagrams symbols, flows and labels must be readable.

Figure 7 Process Flow – Module N

Present in this section all the process flow of the system (users vis-à-vis system modules) using Activity Diagram with Swim Lanes (MWA – both web and mobile system; MAA – either web or mobile system, if applicable otherwise MAA required designs). Describe each illustration explaining the various processes involved in the specific module (Use Style: Capstone Content – Body Text). Note: Diagrams symbols, flows and labels must be readable.

4.3 DESCRIPTION OF THE SYSTEM

In this section of your capstone manuscript, you should include illustration of all major screen designs (MWA – both Web and Mobile Systems; MAA – Storyboard for Film or Animation; MAA System Development – either Web or Mobile Systems) and should contain a brief description of its functions and features. All other screen design or layout should be in the Appendix (Screen Layouts) section of the manuscript with proper label.

Figure 8 Web Application System – Module A

Describe each individual screen designs (module) by explaining the main functions and features. Repeat for other major screen designs (Use Style: Capstone Content – Body Text).

Figure 9 Web Application System – Module N

Describe each individual screen designs (module) by explaining the main functions and features. Repeat for other major screen designs (Use Style: Capstone Content – Body Text).

REFERENCES

In this section, use appropriate referencing tools such as Mendeley or Zotero. Use IEEE refencing styles accordingly (refer to this link https://ieee-dataport.org/sites/default/files/analysis/27/IEEE%20Citation%20Guidelines.pdf as a guide).

APPENDICES

APPENDIX A.
LETTERS AND DOCUMENTATIONS

In this section of your Appendix, scanned clear copy and/or actual document must be included. Note: Documents that should have signatory fields must be completely signed (wet mark is a MUST). Document should be available are as follow(s):

- *[Form] Capstone Team Assignment*
- *[Forms] Request Letter for Client Interview with Non-Disclosure Agreement*
- *[Forms] Request Letter for Capstone Adviser (Letter and Advising Contract)*

APPENDIX B. TRANSCRIPT OF INTERVIEWS AND PHOTO DOCUMENTATION

In this section of your Appendix, scanned clear copy and/or actual document must be included. Note: Documents that should have signatory fields must be completely signed (wet mark is a MUST). Document should be available are as follow(s):

- *Transcript of Interview (All conducted interviews and visits)*
- *Photo Documentation (Proof of Client Interview, Visitation and Consultations)*

APPENDIX C.
PANEL'S LIST OF RECOMMENDATIONS

In this section of your Appendix, scanned clear copy and/or actual document must be included. Note: Documents that should have signatory fields must be completely signed (wet mark is a MUST). Document should be available are as follow(s):

- *Capstone Project 0*
    - *[Forms] Panel's List of Recommendations*